# Media and Cognition
# 媒体与认知

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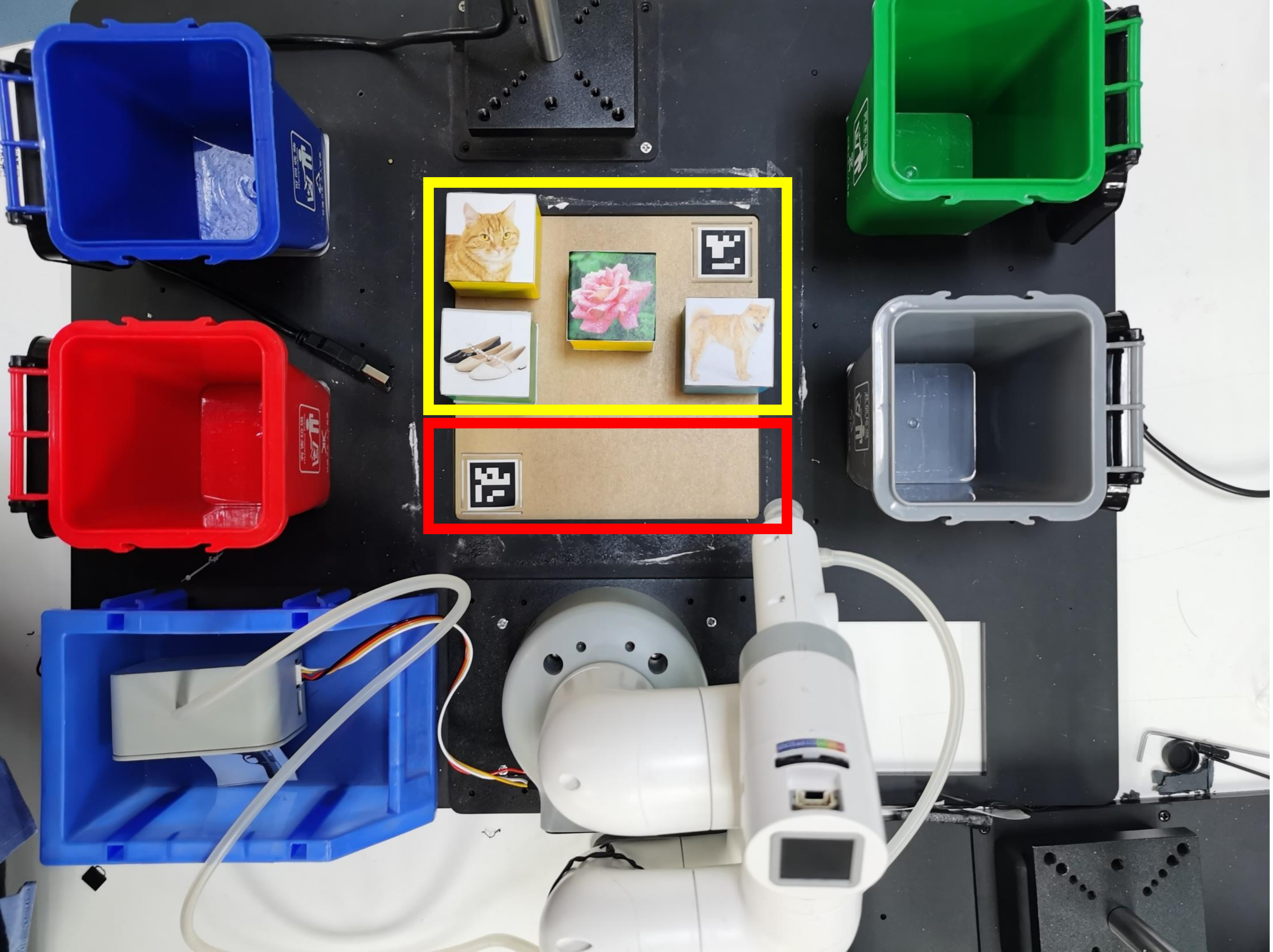
# Lecture 12: Lab Session

## RAP: interactive robotic bin Picking

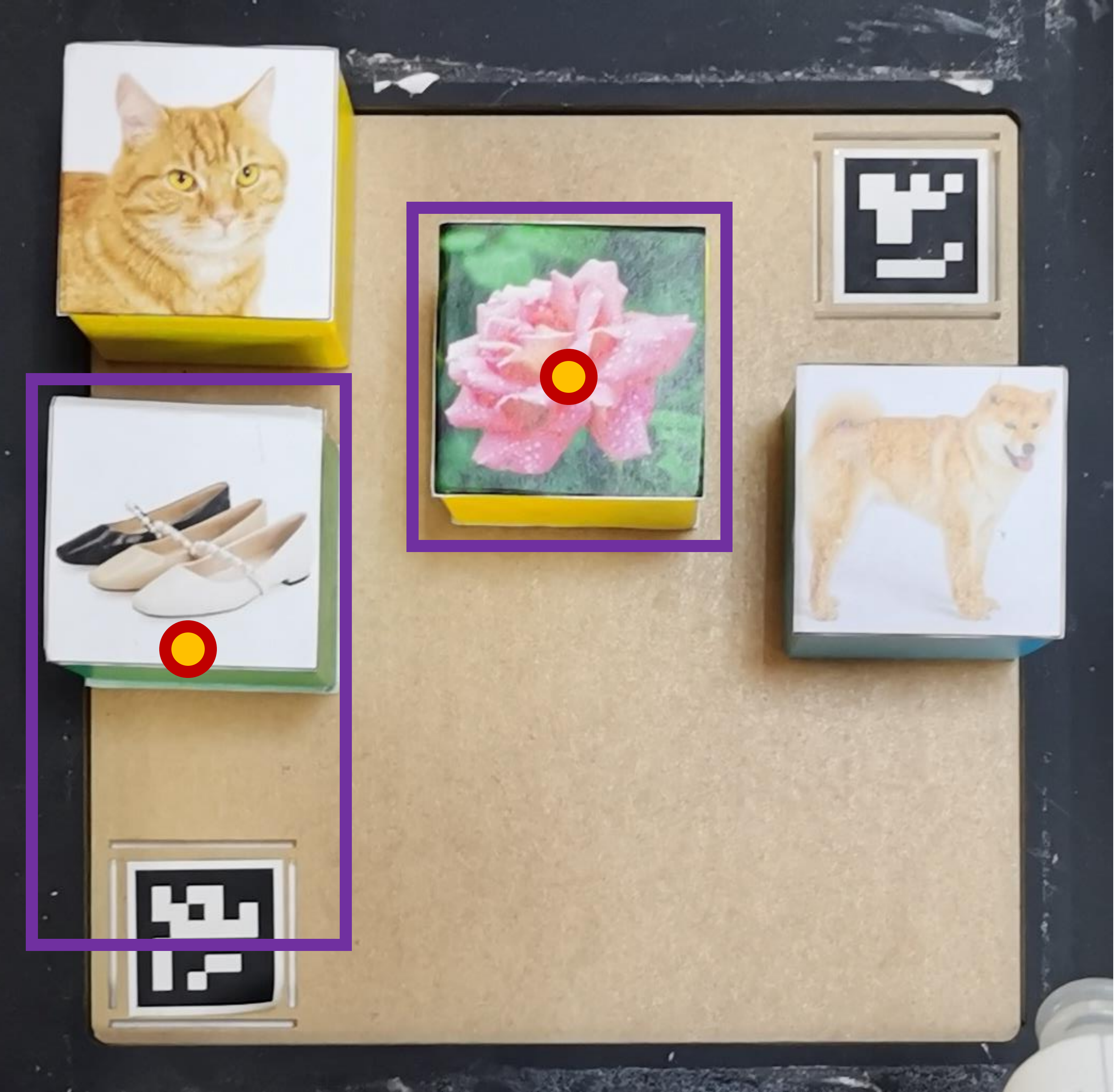
A challenging problem in machine intelligence. Have a **robotic arm** with RGB camera to detect and pick-up **open-vocabulary** targets out of a bin under **natural language instructions**.



# System Overview

Targets should be placed in the yellow region and

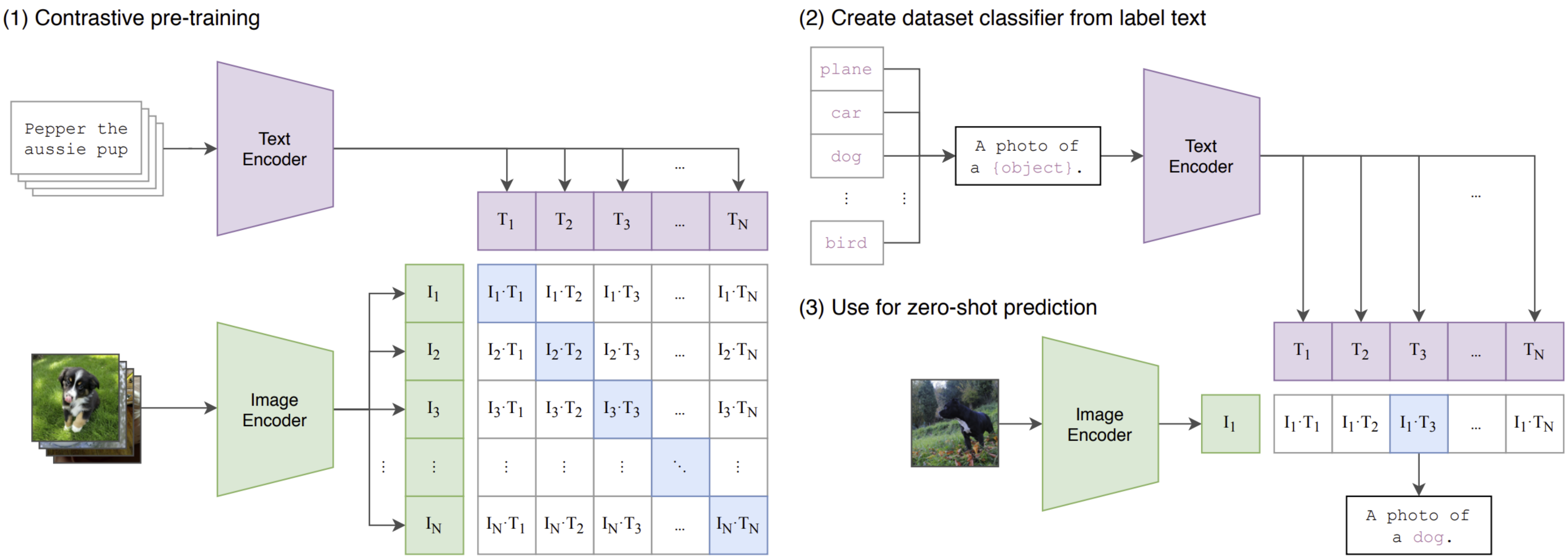
DO NOT place targets in the red region! (robotic arm has difficulty parsing poses in this close-up area)



# Detection

- **Accurate bounding box**
  - Get the ‘center’ coordinate
  - Ensure a stable pick-up
- **Correct recognition**
  - Open-vocabulary: deal with unknown test classes
- **Difficulties:**
  - Domain gap
  - Imaging quality
    - specularity
    - limited resolution

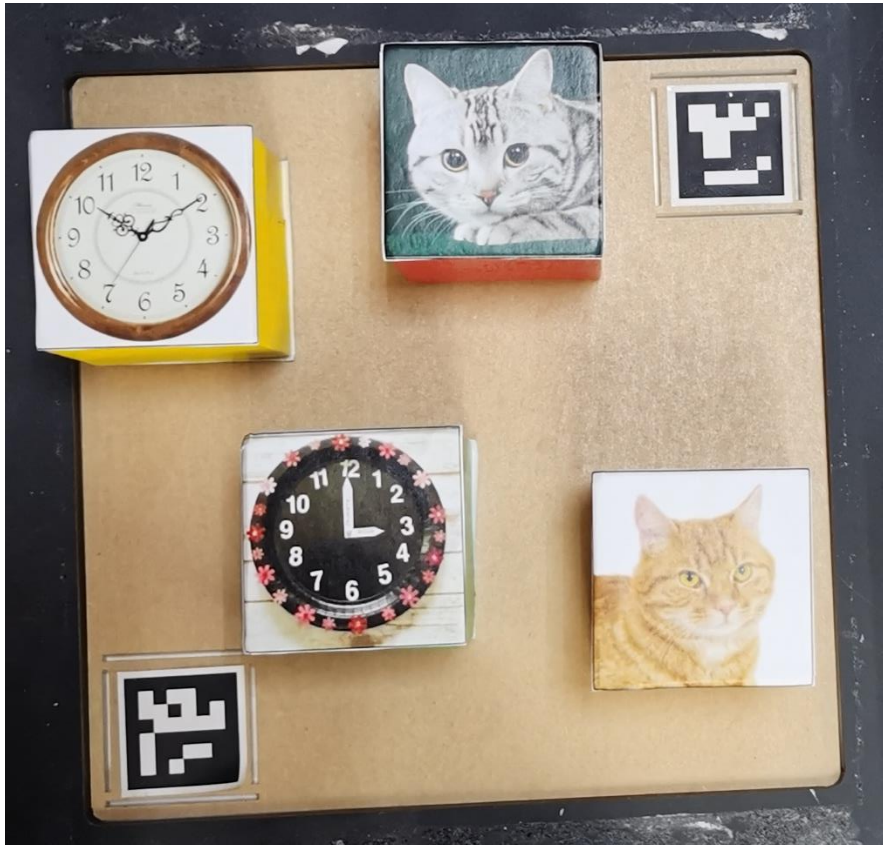
# CLIP (Contrastive Language–Image Pre-training): Connecting text and images




https://openai.com/research/clip https://huggingface.co/docs/transformers/model_doc/clip

Radford, Alec, et al. "Learning transferable visual models from natural language supervision." International conference on machine learning. PMLR, 2021.

# What CLIP can do?



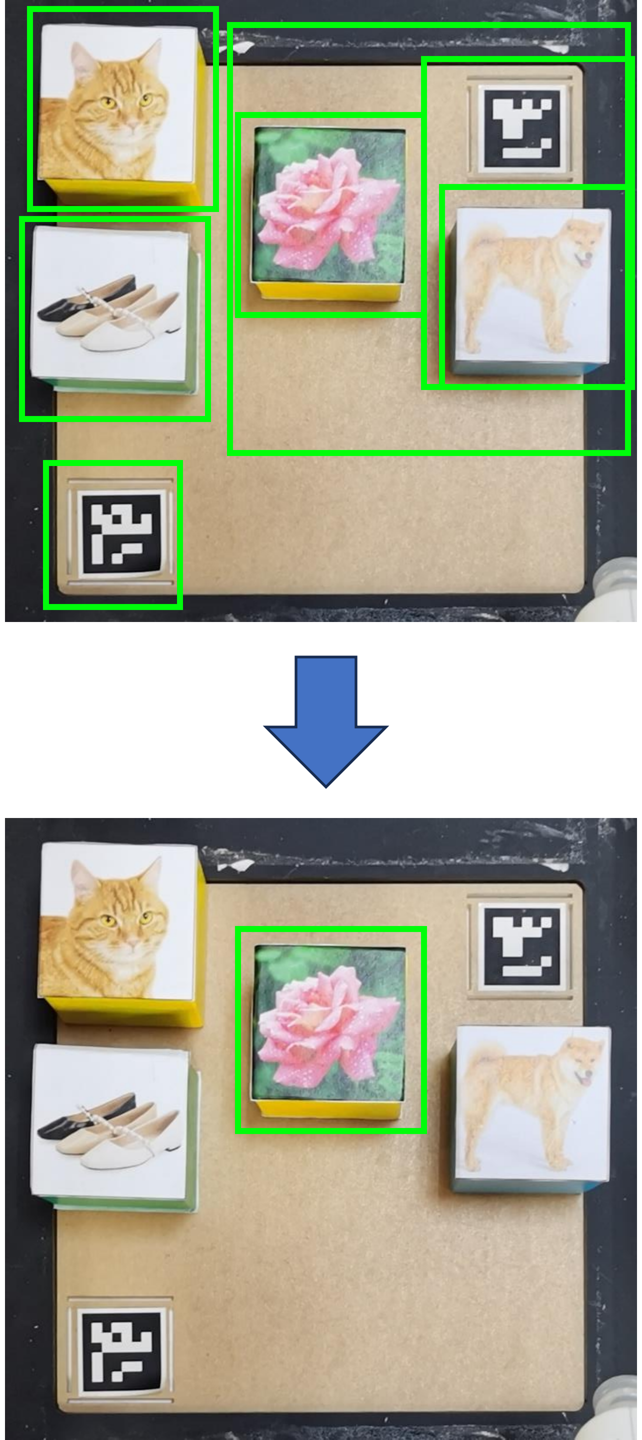
A photo of a **white** cat
A photo of a **yellow** cat
A photo of a **black** clock
A photo of a **white** clock

A photo of a **Corgi**
A photo of a **husky**
A photo of a **labrador**

**Test Criteria:**

- Mean accuracy across N tests
- Reasonable efficiency
- Pick-up a single unique target
- Novel unseen targets for testing
- Text prompt as input, e.g.:
  - a photo of a dog
  - a photo of a red flower
- Prompt engineering is allowed
  - dog -> yellow dog
  - ......
- Targets are fixed-size cubes
- Oriented placement of cubes
- Safe prompt: human / person

**Bonus: additional improve-ments from various aspects are well appreciated**



**Some Hints:**

**Deploying pretrained model is enough**

- CLIP inference can be done on CPUs
- Finetuning on our case is not recommended

**One plausible solution:**
**Building a 2-stage open vocabulary detector**

- Region proposal with geometric cues
- Pretrained CLIP as classifier

**Another plausible solution:**
**Advanced open-vocabulary detectors**

- Pay attention to the robustness towards our case
  - Region proposal may not be robust
  - CLIP finetuned on downstream datasets are often not robust enough

# Grading Criteria *

**Stage 1: Trans-Modality Media (Detection)**: 50% in total

1. Accuracy:
    1. Detection bounding boxes: 20%
    2. Prompts translation & understanding: 20%
2. Coding accessibility: 5%
3. Bonus: 5% (algorithm novelty, additional functionality, additional discoveries, …)

**Stage 2: Cognition-Based Execution (Robotic Arm)**: 50% in total

1. System design: 5%
2. Accuracy and efficiency:
    1. Recognition: 10%
    2. Localization: 10%
    3. Operation: 10%
3. Coding accessibility: 5%
4. Bonus: 10% (extreme cases, occlusion, interruption, …)

*Work in **teams of 3 to 4 students**.*

*End-of-term presentations and assessments.*

* Note: This criteria may vary according to your practice.

# *Good luck on the grand tour!*

- *Q & A*